

Impact of Nrf2 on UVB-induced skin inflammation/photoprotection and photoprotective effect of sulforaphane.



Ultraviolet (UV) of sunlight is a complete carcinogen that can burn skin, enhance inflammation, and drive skin carcinogenesis. Previously, we have shown that sulforaphane (SFN) inhibited chemically induced skin carcinogenesis via nuclear factor (erythroid-derived 2)-like 2 (Nrf2) and others have shown that broccoli sprout extracts containing high SFN protected against UV-induced skin carcinogenesis in SKH-1 hairless mice. A recent study showed that there was no difference between Nrf2 knockout (Nrf2 KO) and Nrf2 wild-type (WT) BALB/C mice after exposing to high dose of UVB. Since Nrf2 plays critical roles in the anti-oxidative stress/anti-inflammatory responses, it is relevant to assess the role of Nrf2 for photoprotection against UV. In this context, the role of Nrf2 in UVB-induced skin inflammation in Nrf2 WT and Nrf2 KO C57BL/6 mice was studied. A single dose of UVB (300 mJ/cm²) resulted in skin inflammation in both WT and Nrf2 KO (-/-) mice (KO mice) at 8 h and 8 d following UVB irradiation. In the WT mice inflammation returned to the basal level to a greater extent when compared to the KO mice. SFN treatment of Nrf2 WT but not Nrf2 KO mice restored the number of sunburn cells back to their basal level by 8 d after UVB irradiation. Additionally, UVB-induced short-term inflammatory biomarkers (interleukin-1 β and interleukin-6) were increased in the KO mice and UVB-induced apoptotic cells in the KO mice were significantly higher as compared to that in the WT. Taken together, our results show that functional Nrf2 confers a protective effect against UVB-induced inflammation, sunburn reaction, and SFN-mediated photoprotective effects in the skin. Copyright © 2010 Wiley-Liss, Inc.